Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (C6H6)
Basis set   : cc-pVDZ
Charge   : 0
Multiplicity : 1
Total e-   : 42
n-body order : 4

GROUND STATE ENERGIES

E0 (Ha) : -230.75355805
E1 (Ha) : -230.57826157
Gap (Ha) : 0.17529648
HF (Ha) : -230.71878874

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 3

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 18
Frozen e-   : 36
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.961506
<D> / site : 0.493584

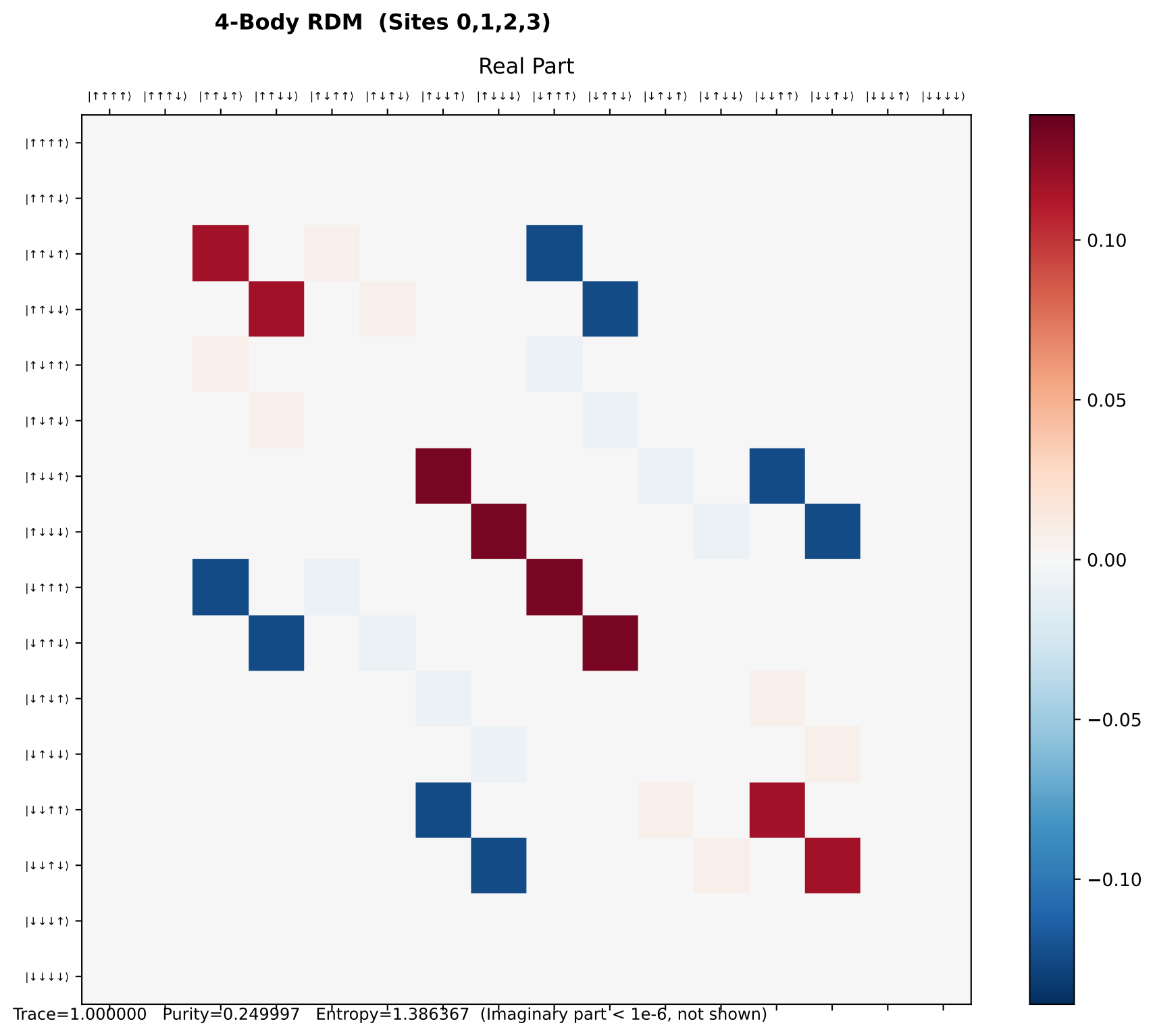
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Body order : 4
Clusters   : 15

Date: 2026-07-10 17:23

Summary — 4-Body RDMs (Benzene, CAS(6,6))

Orbitals	Purity	C_GME (pure)	C_LB	C_gcg	C_stiefel	C_UB	State
0,1,2,3	0.249997	—	—	—	—	—	MIXED
0,1,2,4	0.249997	—	—	—	—	—	MIXED
0,1,2,5	0.834591	—	—	—	—	—	WEAKLY MIXED
0,1,3,4	0.249998	—	—	—	—	—	MIXED
0,1,3,5	0.228267	—	—	—	—	—	MIXED
0,1,4,5	0.228210	—	—	—	—	—	MIXED
0,2,3,4	0.249998	—	—	—	—	—	MIXED
0,2,3,5	0.228511	—	—	—	—	—	MIXED
0,2,4,5	0.228456	—	—	—	—	—	MIXED
0,3,4,5	0.801534	—	—	—	—	—	WEAKLY MIXED
1,2,3,4	1.000000	—	—	—	—	—	PURE
1,2,3,5	0.192480	—	—	—	—	—	STRONGLY MIXED
1,2,4,5	0.191842	—	—	—	—	—	STRONGLY MIXED
1,3,4,5	0.212280	—	—	—	—	—	MIXED
2,3,4,5	0.212777	—	—	—	—	—	MIXED


4-Body RDM (Sites 0,1,2,3)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249997 Entropy=1.386367 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,3)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2499972350
Von Neumann entropy	: 1.3863667067
Eigenvalue min	: -1.280478e-17
Eigenvalue max	: 2.499991e-01
Imaginary part	: < 1e-6 (not shown)

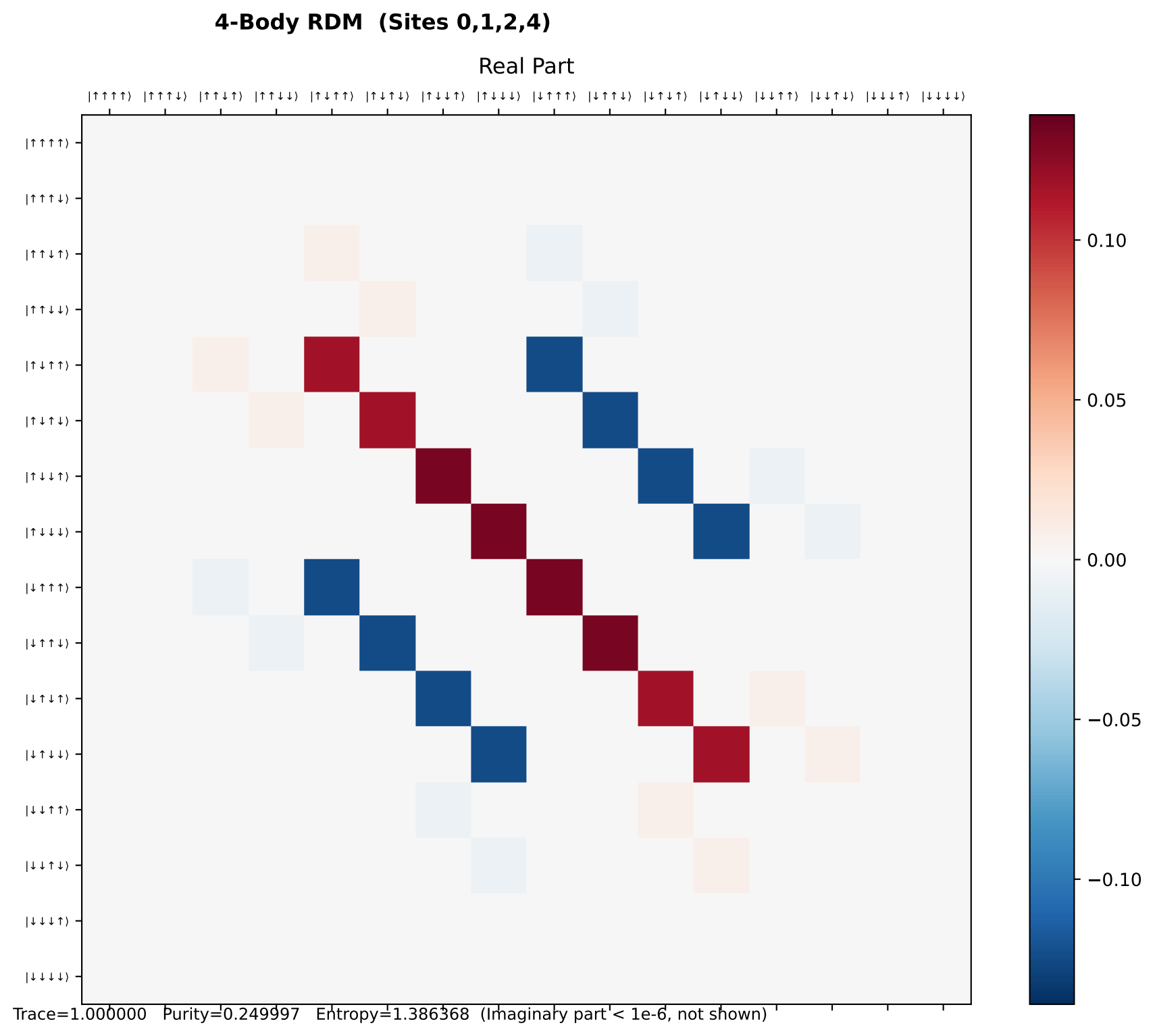
GME Analysis
==========

gamma = Tr(rho^2) = 0.24999724

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,1,2,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249997 Entropy=1.386368 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2499971871
Von Neumann entropy	1.3863678625
Eigenvalue min	-5.083853e-18
Eigenvalue max	2.499991e-01
Imaginary part	< 1e-6 (not shown)

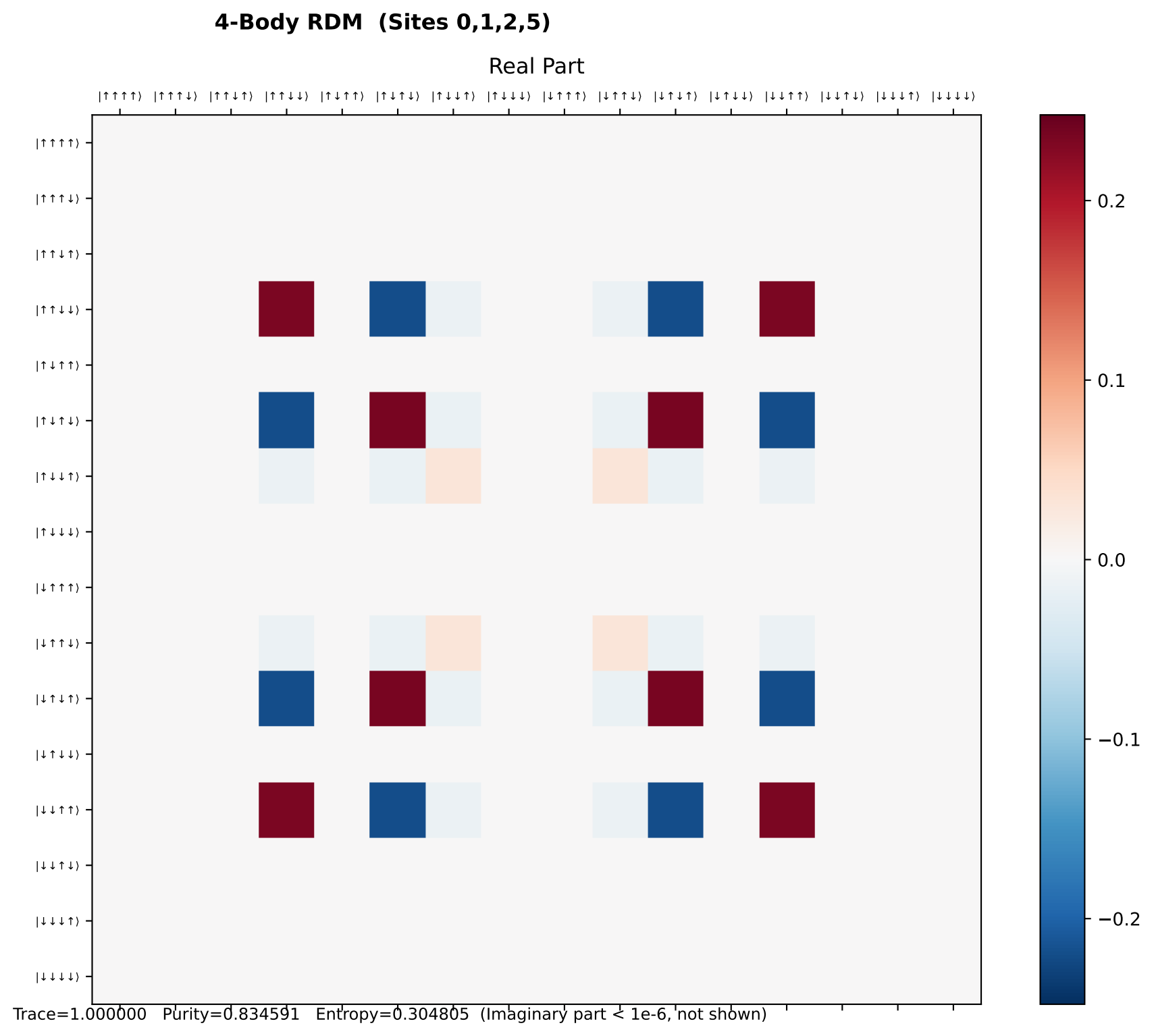
GME Analysis
==========

gamma = Tr(rho^2) = 0.24999719

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,1,2,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.834591 Entropy=0.304805 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,2,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.8345909728
Von Neumann entropy	: 0.3048046140
Eigenvalue min	: -3.205555e-17
Eigenvalue max	: 9.090177e-01
Imaginary part	: < 1e-6 (not shown)

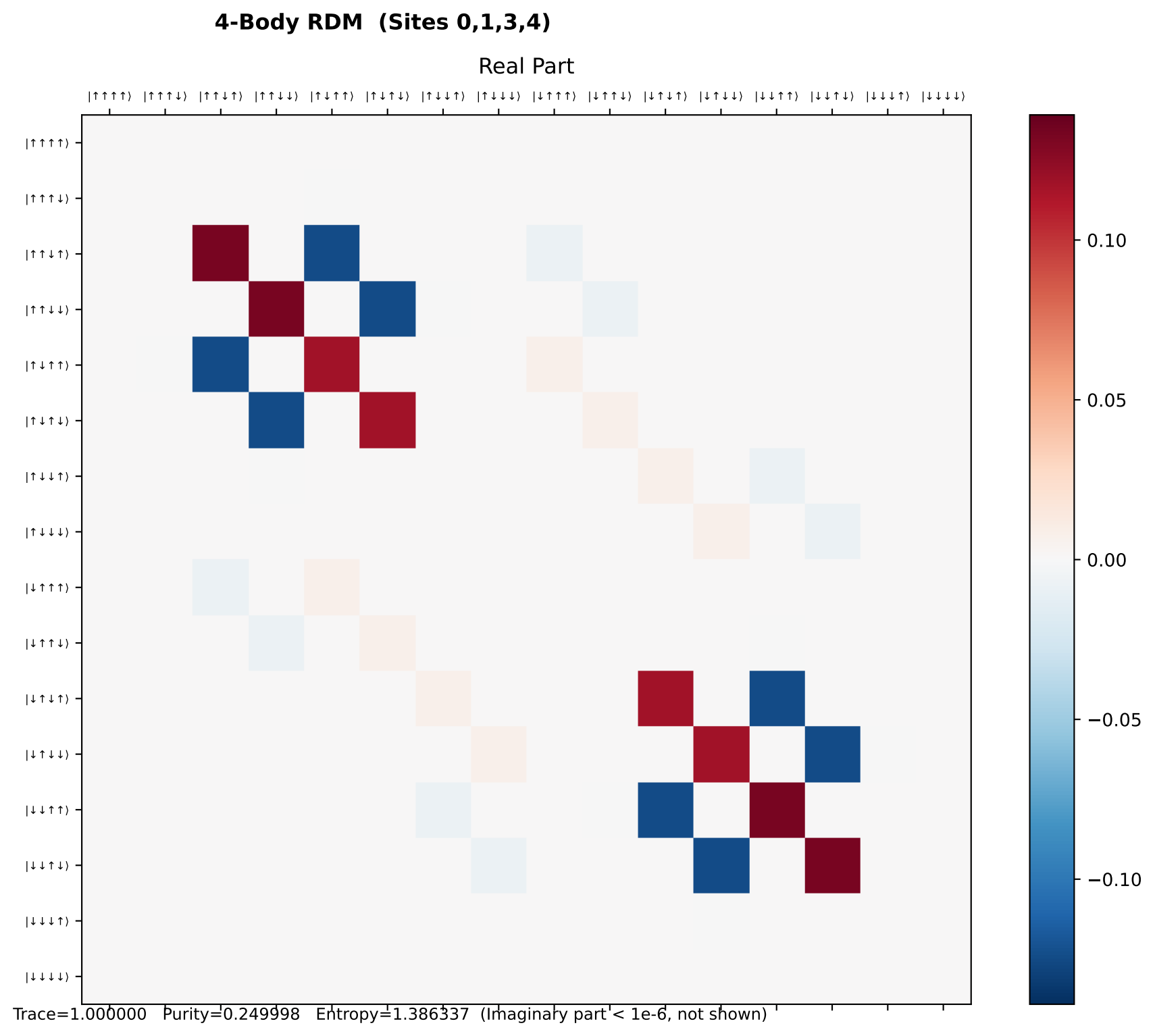
GME Analysis
==========

gamma = Tr(rho^2) = 0.83459097

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,1,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249998 Entropy=1.386337 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.2499984586
Von Neumann entropy	1.3863365325
Eigenvalue min	-2.695504e-17
Eigenvalue max	2.499993e-01
Imaginary part	< 1e-6 (not shown)

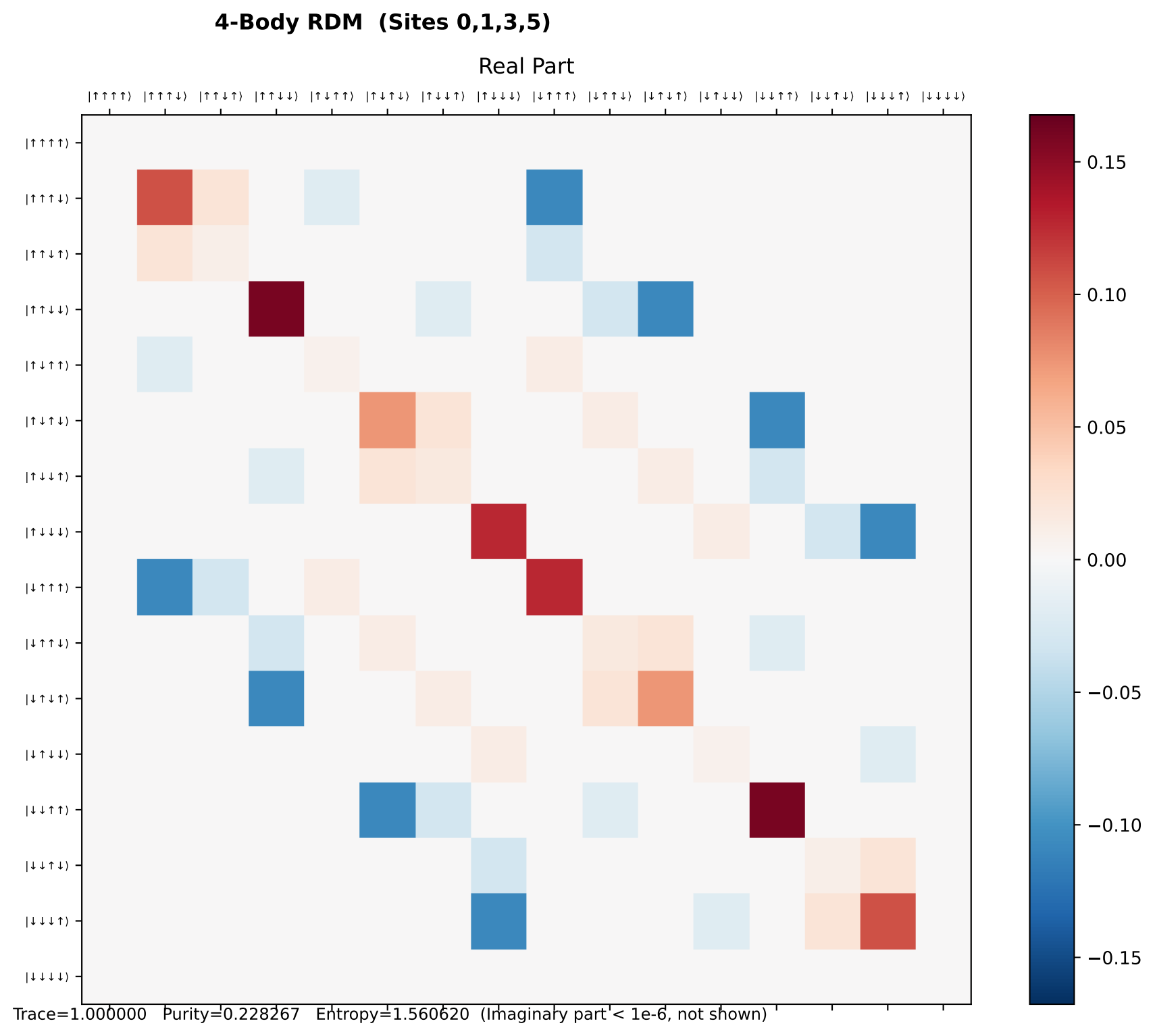
GME Analysis
==========

gamma = Tr(rho^2) = 0.24999846

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,1,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228267 Entropy=1.560620 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,3,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2282670771
Von Neumann entropy	: 1.5606201050
Eigenvalue min	: -3.523871e-17
Eigenvalue max	: 2.499215e-01
Imaginary part	: < 1e-6 (not shown)

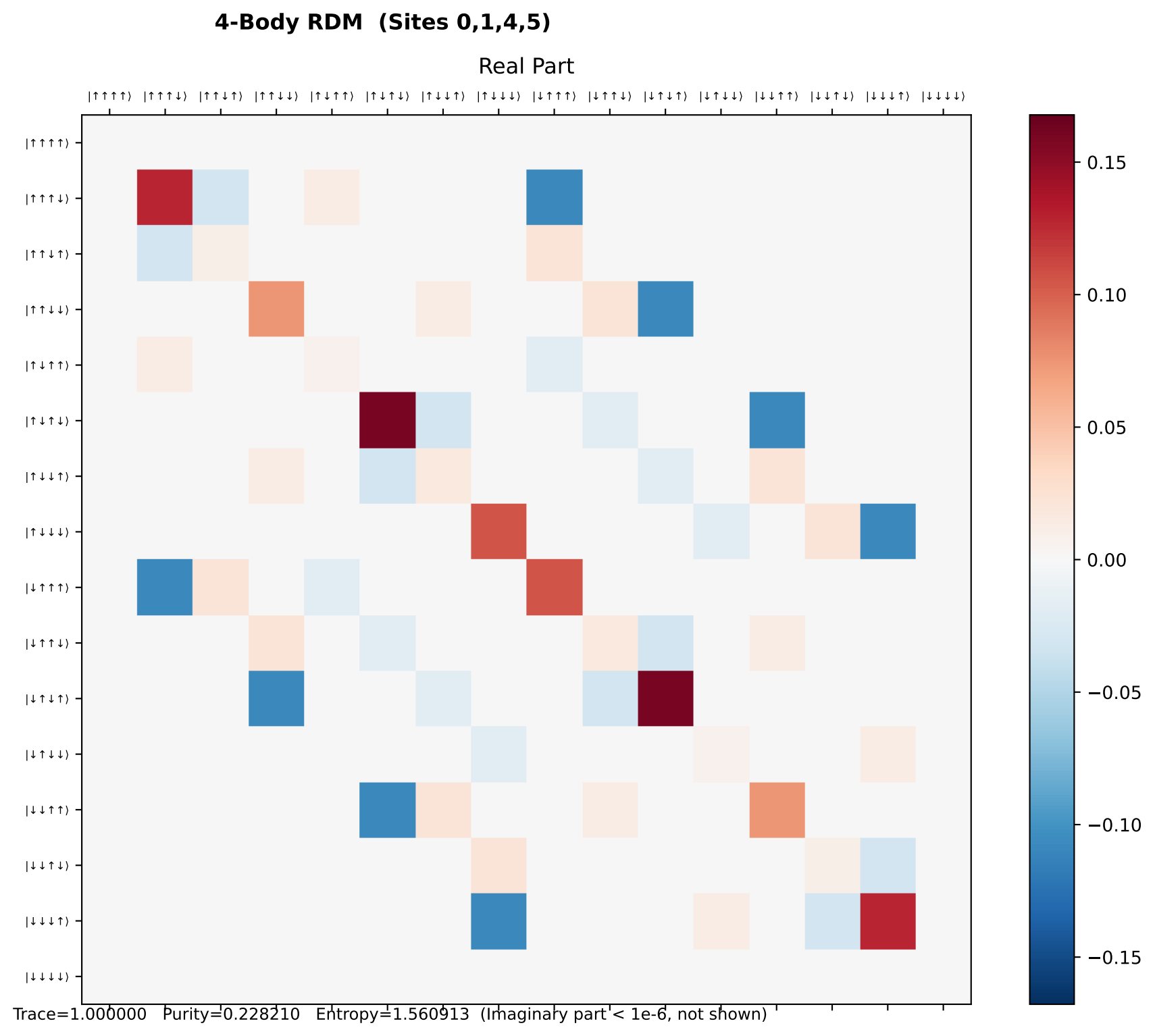
GME Analysis
==========

gamma = Tr(rho^2) = 0.22826708

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,1,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228210 Entropy=1.560913 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,1,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2282097687
Von Neumann entropy	: 1.5609134306
Eigenvalue min	: -4.029277e-17
Eigenvalue max	: 2.499339e-01
Imaginary part	: < 1e-6 (not shown)

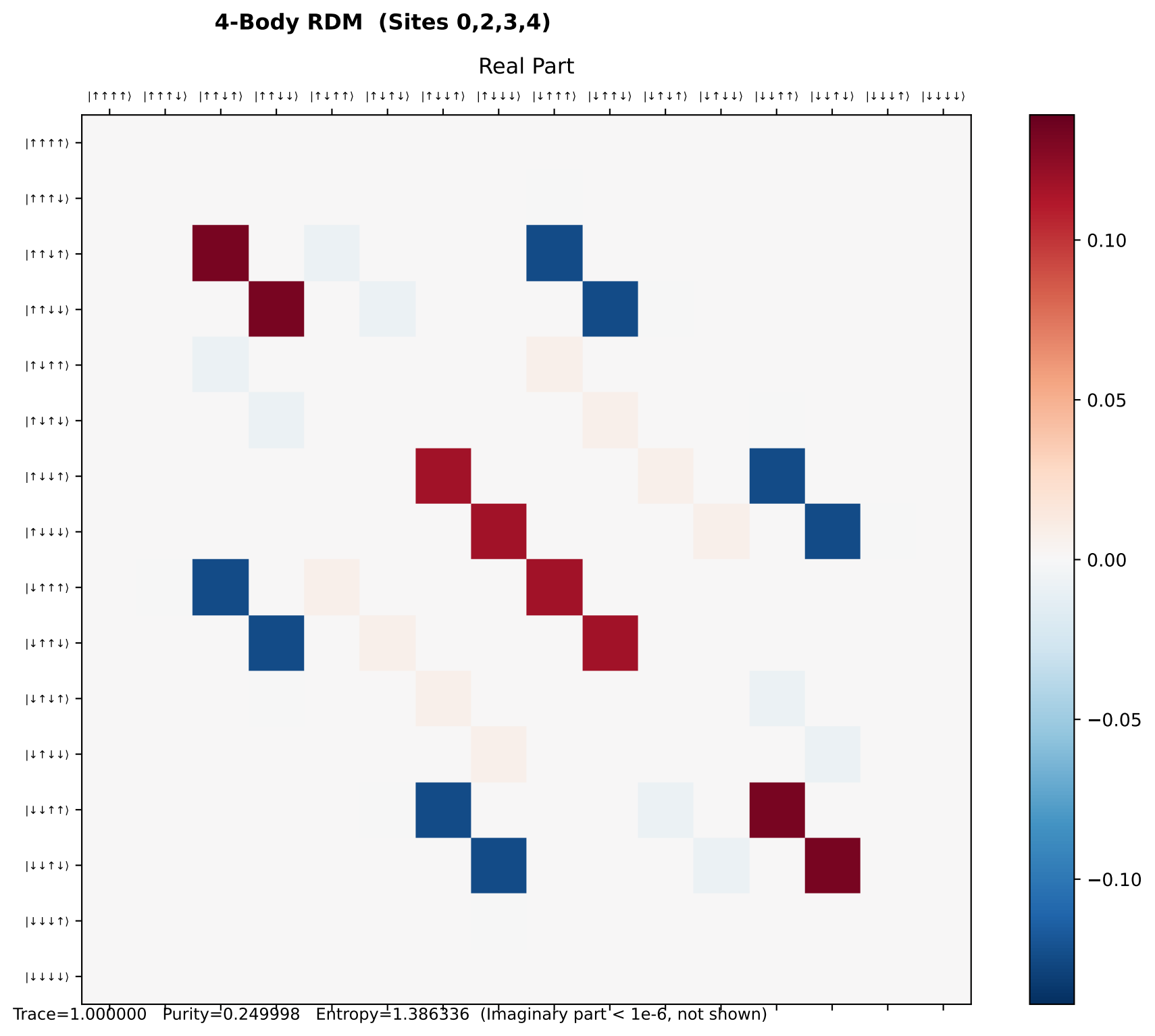
GME Analysis
==========

gamma = Tr(rho^2) = 0.22820977

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,2,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.249998 Entropy=1.386336 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,2,3,4)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2499984844
Von Neumann entropy	: 1.3863358790
Eigenvalue min	: -2.564762e-17
Eigenvalue max	: 2.499993e-01
Imaginary part	: < 1e-6 (not shown)

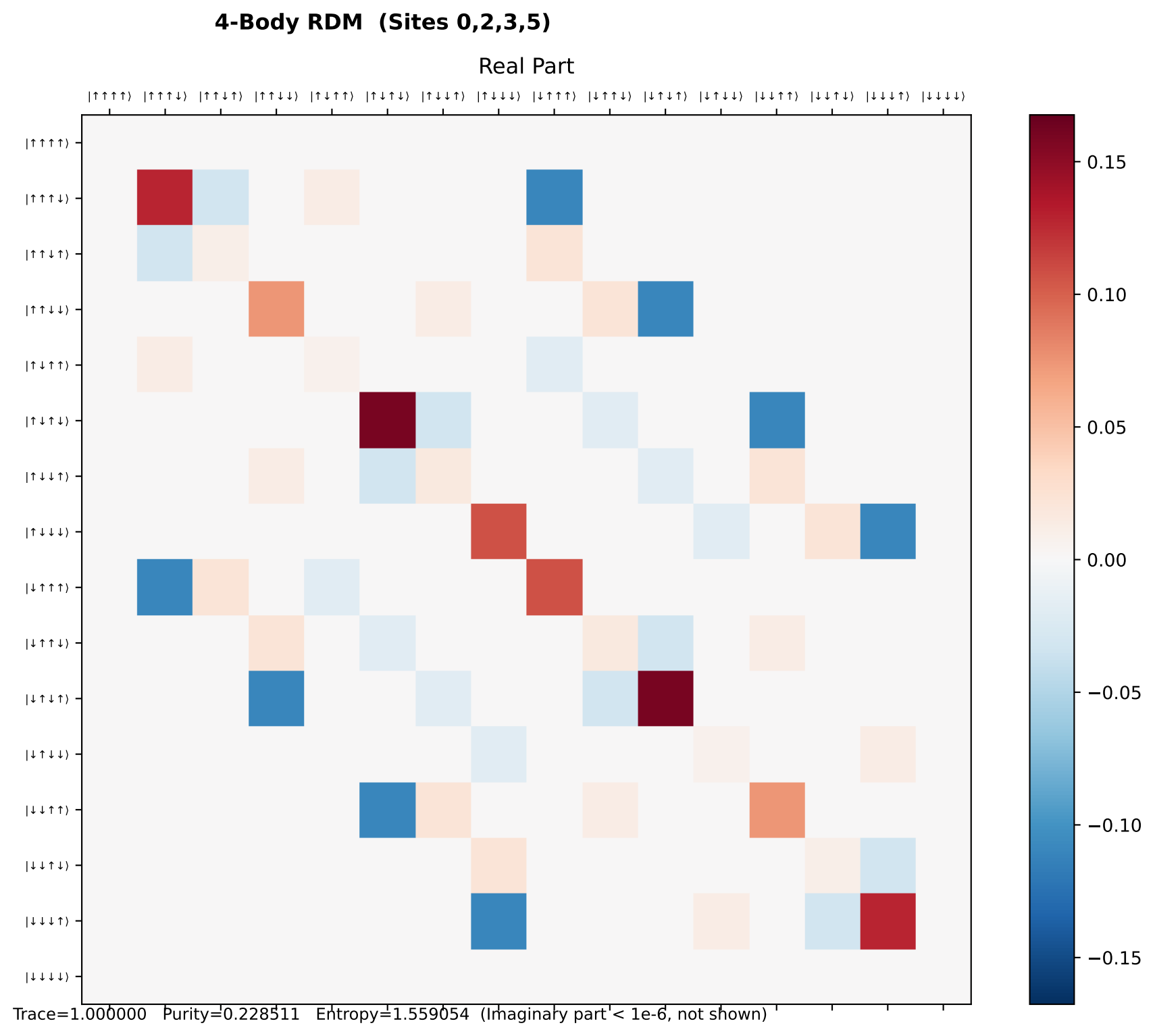
GME Analysis
==========

gamma = Tr(rho^2) = 0.24999848

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,2,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228511 Entropy=1.559054 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,2,3,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2285113709
Von Neumann entropy	: 1.5590535668
Eigenvalue min	: -2.557186e-17
Eigenvalue max	: 2.499308e-01
Imaginary part	: < 1e-6 (not shown)

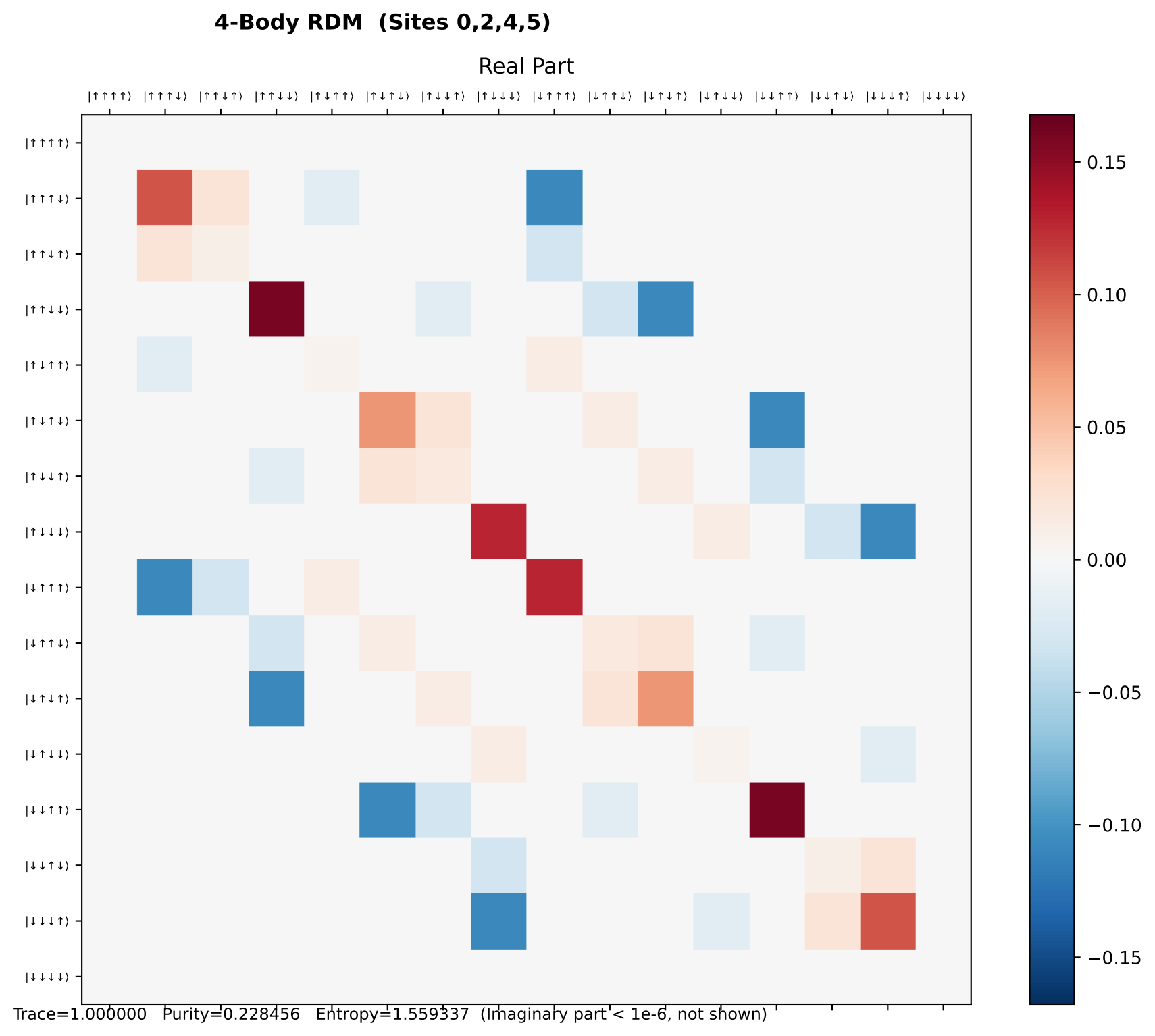
GME Analysis
==========

gamma = Tr(rho^2) = 0.22851137

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,2,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
Trace=1.000000 Purity=0.228456 Entropy=1.559337 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,2,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2284561949
Von Neumann entropy	: 1.5593374628
Eigenvalue min	: -4.007288e-17
Eigenvalue max	: 2.499424e-01
Imaginary part	: < 1e-6 (not shown)

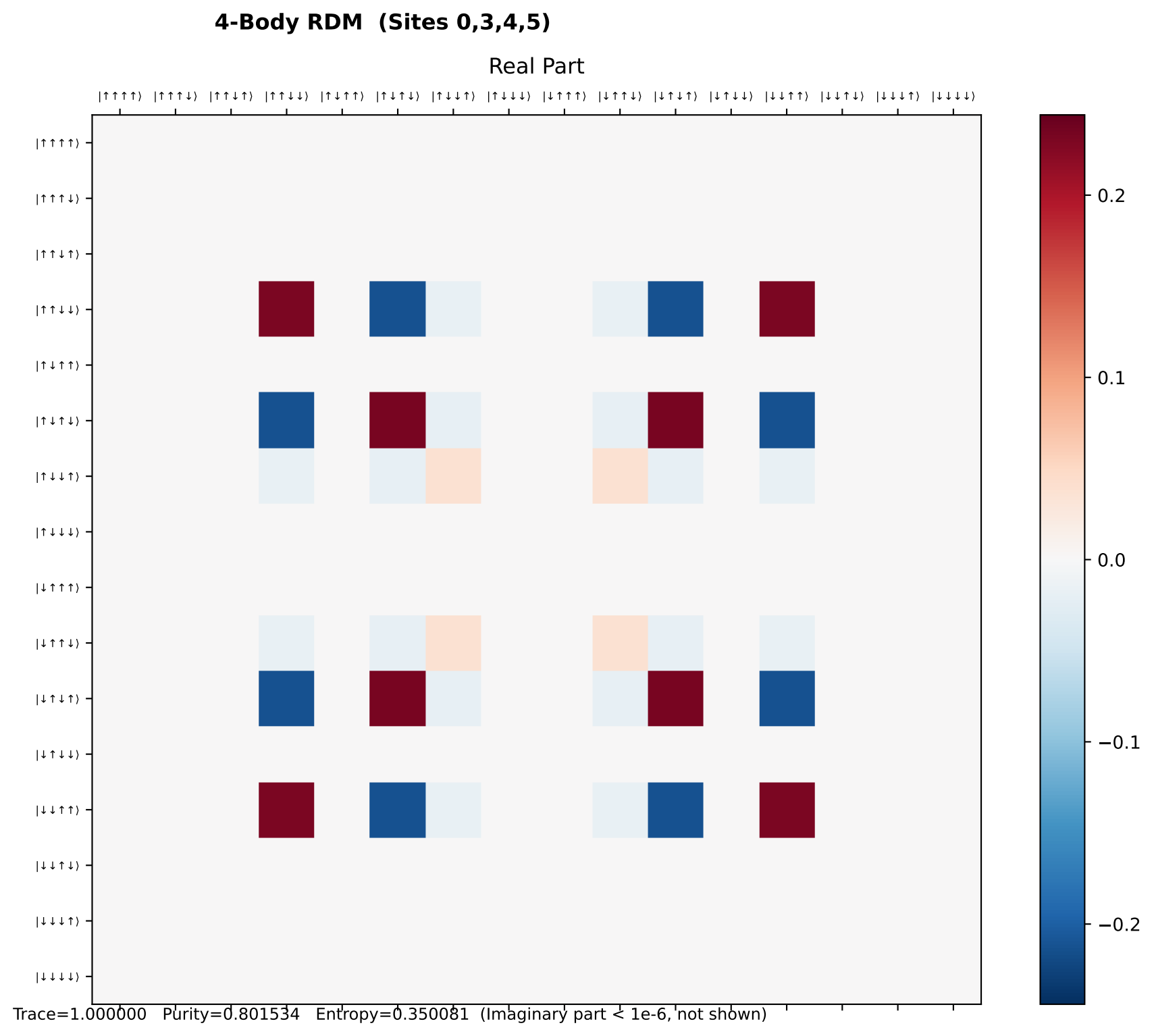
GME Analysis
==========

gamma = Tr(rho^2) = 0.22845619

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 0,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=0.801534 Entropy=0.350081 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 0,3,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.8015339801
Von Neumann entropy	: 0.3500813350
Eigenvalue min	: -3.997935e-17
Eigenvalue max	: 8.882873e-01
Imaginary part	: < 1e-6 (not shown)

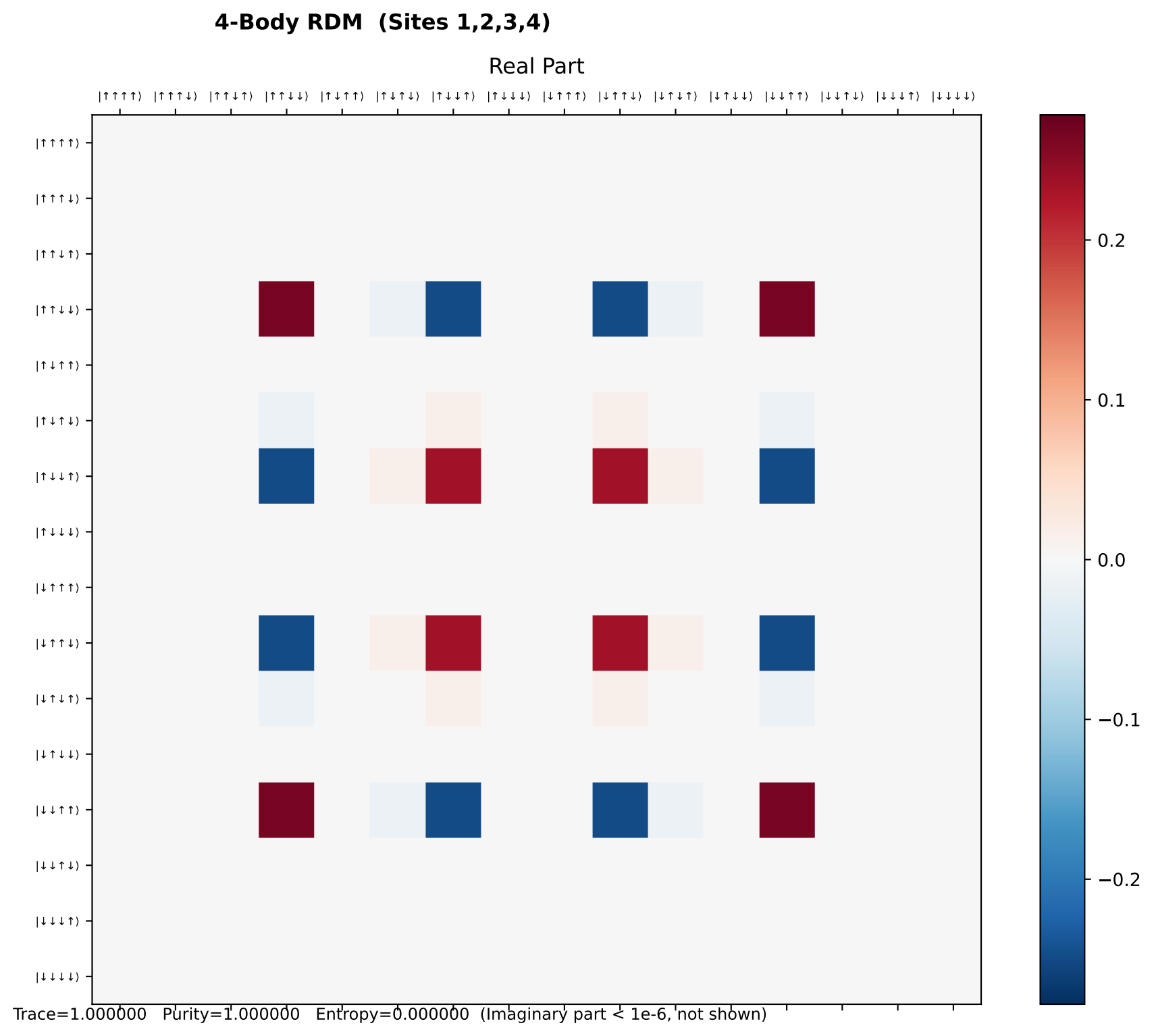
GME Analysis
==========

gamma = Tr(rho^2) = 0.80153398

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 1,2,3,4)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.2
0.1
0.0
−0.1
−0.2
Trace=1.000000 Purity=1.000000 Entropy=0.000000 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,4)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.9999999998
Von Neumann entropy	0.0000000019
Eigenvalue min	-1.306012e-16
Eigenvalue max	1.000000e+00
Imaginary part	< 1e-6 (not shown)

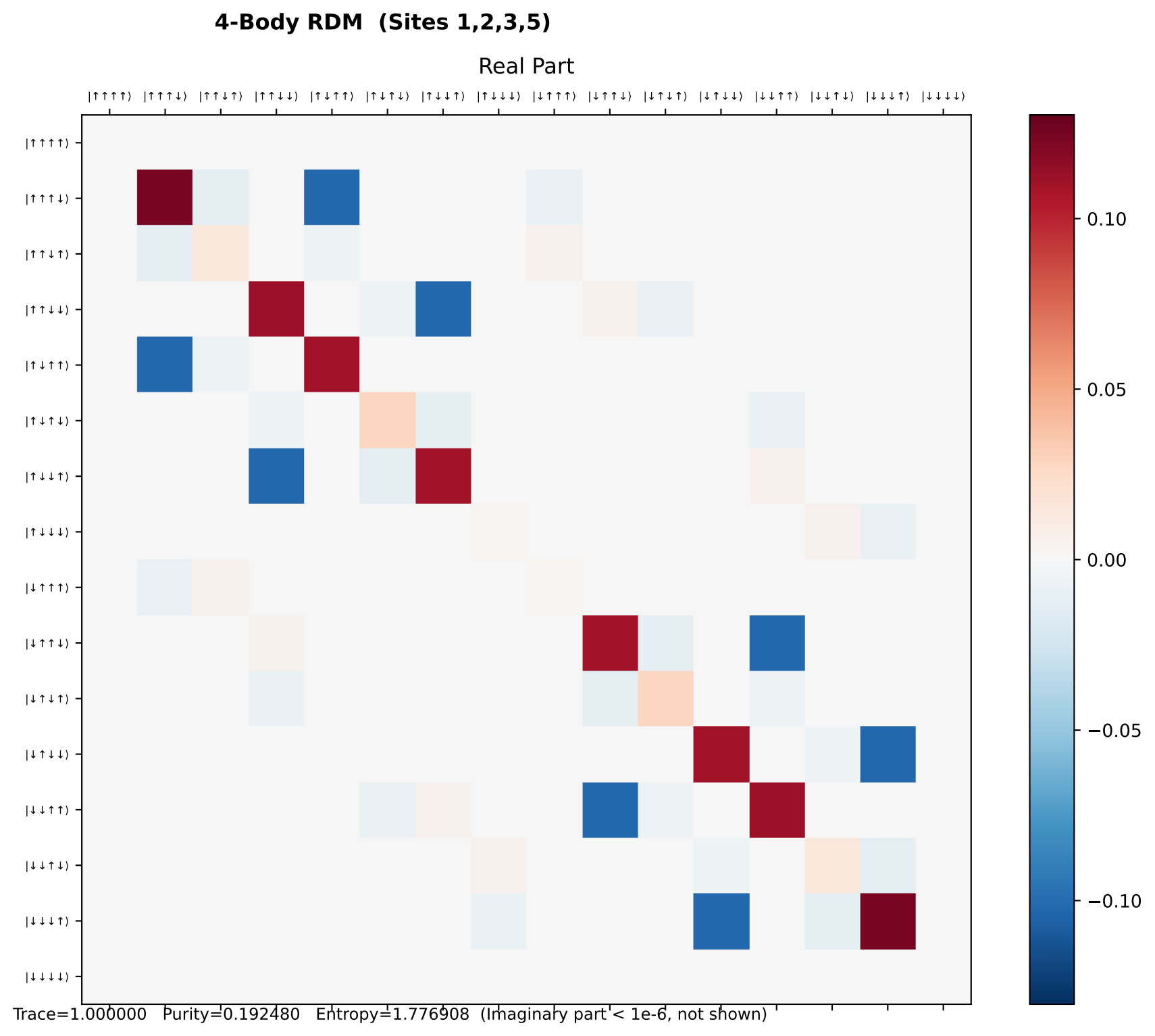
GME Analysis
==========

gamma = Tr(rho^2) = 1.00000000

A	B	Tr(rho_A^2)
(1,)	(2, 3, 4)	0.50000000
(1, 2)	(3, 4)	0.25271107
(1, 3)	(2, 4)	0.99421839
(1, 4)	(2, 3)	0.25307054
(1, 2, 3)	(4,)	0.50000000
(1, 2, 4)	(3,)	0.50000000
(1, 3, 4)	(2,)	0.50000000

STATUS: GME EXISTS

C_GME = 0.10753238
GME confirmed across ALL cuts.


4-Body RDM (Sites 1,2,3,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.192480 Entropy=1.776908 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,3,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.1924804145
Von Neumann entropy	: 1.7769084571
Eigenvalue min	: -2.654149e-17
Eigenvalue max	: 2.198107e-01
Imaginary part	: < 1e-6 (not shown)

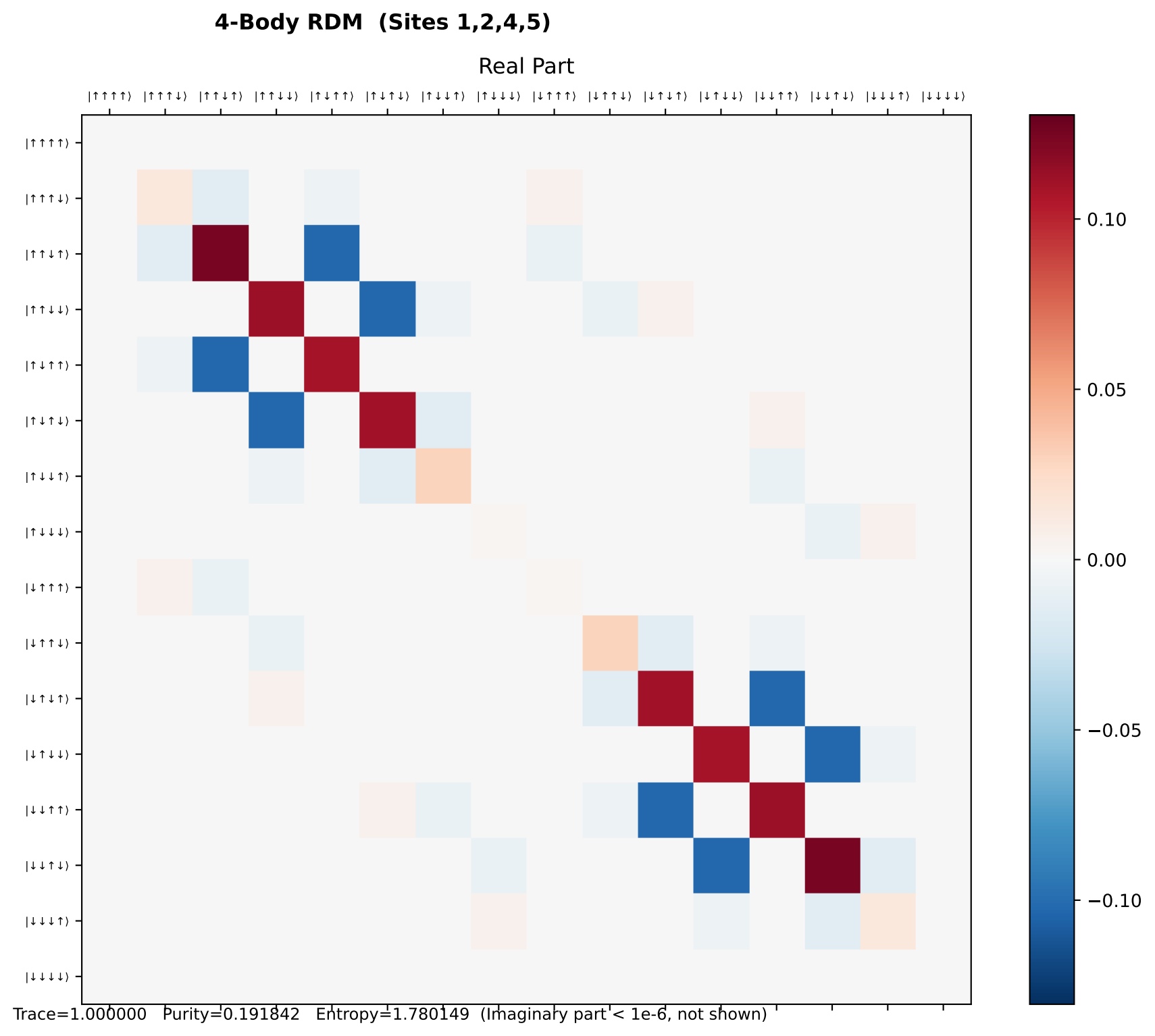
GME Analysis
==========

gamma = Tr(rho^2) = 0.19248041

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 1,2,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.191842 Entropy=1.780149 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,2,4,5)

RDM Metrics
==========

Trace(rho)	1.0000000000
Purity Tr(rho^2)	0.1918423382
Von Neumann entropy	1.7801492269
Eigenvalue min	-4.908832e-17
Eigenvalue max	2.194390e-01
Imaginary part	< 1e-6 (not shown)

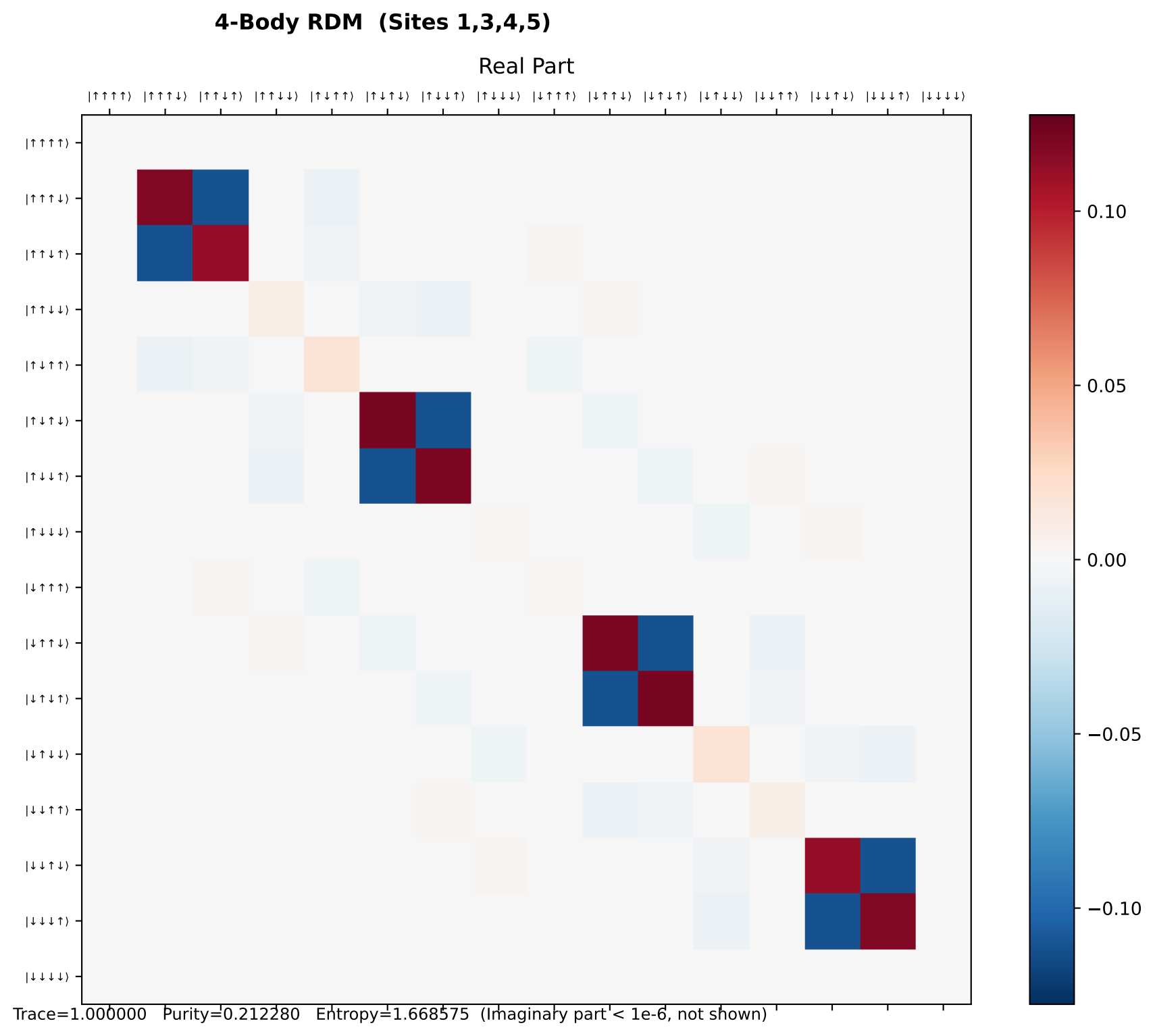
GME Analysis
==========

gamma = Tr(rho^2) = 0.19184234

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 1,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.212280 Entropy=1.668575 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 1,3,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2122800809
Von Neumann entropy	: 1.6685754151
Eigenvalue min	: -2.046097e-17
Eigenvalue max	: 2.380752e-01
Imaginary part	: < 1e-6 (not shown)

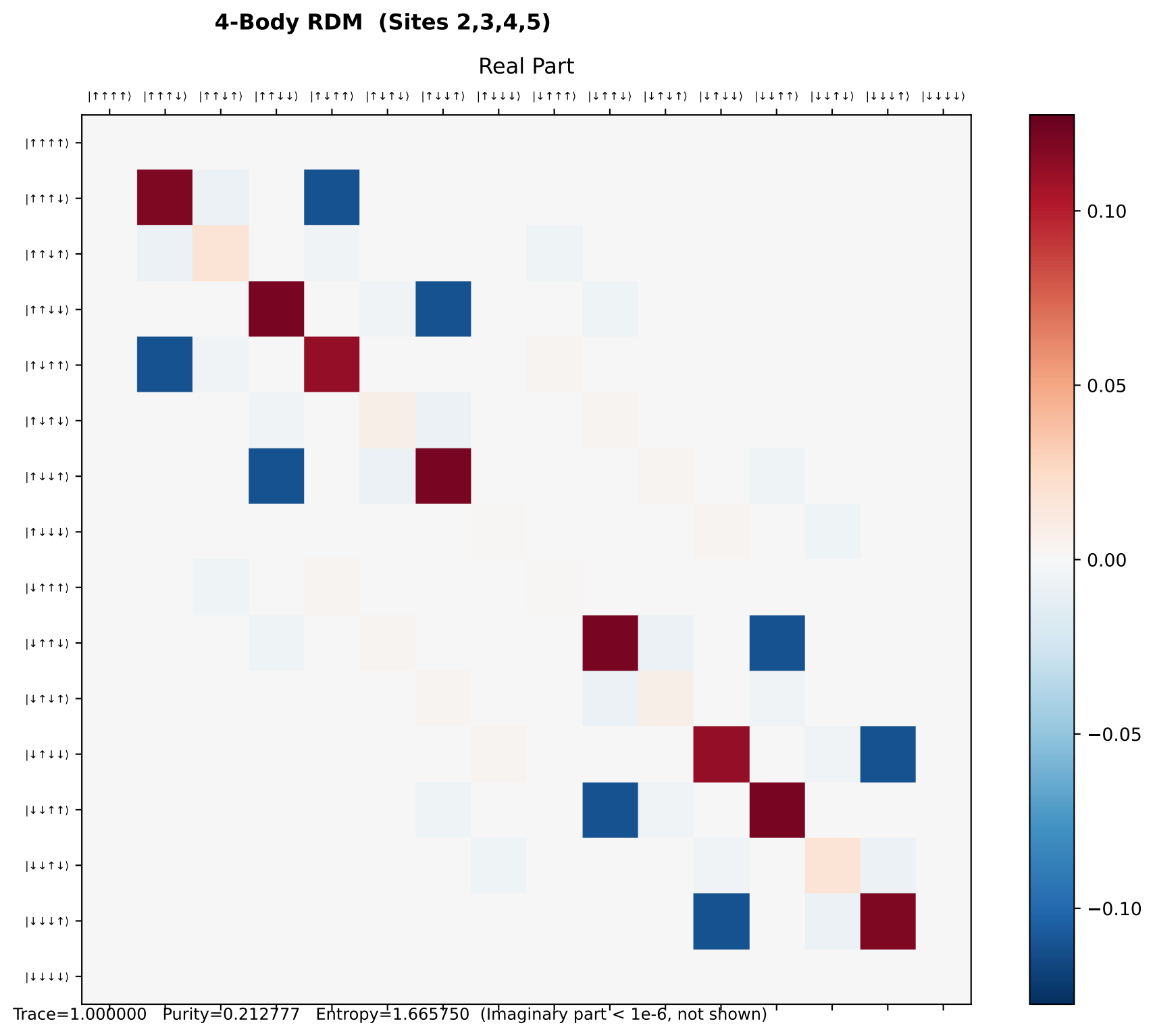
GME Analysis
==========

gamma = Tr(rho^2) = 0.21228008

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.


4-Body RDM (Sites 2,3,4,5)
Real Part
|↑↑↑↑⟩ |↑↑↑↓⟩ |↑↑↓↑⟩ |↑↑↓↓⟩ |↑↓↑↑⟩ |↑↓↑↓⟩ |↑↓↓↑⟩ |↑↓↓↓⟩ |↓↑↑↑⟩ |↓↑↑↓⟩ |↓↑↓↑⟩ |↓↑↓↓⟩ |↓↓↑↑⟩ |↓↓↑↓⟩ |↓↓↓↑⟩ |↓↓↓↓⟩
0.10
0.05
0.00
−0.05
−0.10
Trace=1.000000 Purity=0.212777 Entropy=1.665750 (Imaginary part < 1e-6, not shown)

Analysis Summary - 4-Body RDM (Sites 2,3,4,5)

RDM Metrics
==========

Trace(rho)	: 1.0000000000
Purity Tr(rho^2)	: 0.2127765290
Von Neumann entropy	: 1.6657503887
Eigenvalue min	: -4.338199e-17
Eigenvalue max	: 2.381848e-01
Imaginary part	: < 1e-6 (not shown)

GME Analysis
==========

gamma = Tr(rho^2) = 0.21277653

A	B	Tr(rho_A^2)

STATUS: MIXED STATE

Tr(rho^2) < 1. Pure-state C_GME not applicable. See convex-roof result below.